

Subject: Software Testing	Subject Code: 316314
Semester: 06	Course: CO6K-A
Laboratory No: L003A	Name of Subject Teacher: Mrs. Sangeeta Shrisat
Name of Student: Vedant Atul Shirke	Roll ID: 23203A0018

Experiment No:	07
Title of Experiment:	Validate login procedure for E-Commerce Application (Flipkart or Amazon)

I. Practical Significance:

This practical introduces functional and boundary testing techniques for validating the login process in an E-Commerce Application (such as Flipkart or Amazon). It focuses on ensuring that valid credentials grant access, while invalid, missing, or malformed inputs are handled securely. This practical also helps learners understand how to apply Equivalence Class Partitioning (ECP) and Boundary Value Analysis (BVA) in Black Box Testing for authentication modules.

II. Practical Related Questions:

1. Design five additional test cases for login fields using ECP and BVA.

ANS:-

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_01	(BVA) Verify behavior when Mobile Number is less than 10 digits.	1. Enter Mobile: 987654321 (9 digits) 2. Password: Pass@1233. Click Login	Error message: "Please enter a valid 10-digit mobile number."	Error message: "Please enter a valid 10-digit mobile number."	Pass
TC_02	(BVA) Verify	1. Enter Email:	Error message:	Error message:	Pass



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Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
	behavior when Password is below minimum length (e.g., <8 chars).	shreyas@gmail.com 2. Password: 1234 (4 chars) 3. Click Login	"Password must be at least 8 characters long."	"Password must be at least 8 characters long."	
TC_03	(ECP) Verify login with an unregistered Email address (Valid format, invalid data).	1. Enter Email: shreyas26@gmail.com 2. Password: Pass@1233. Click Login	Error message: "We cannot find an account with that email address."	Error message: "We cannot find an account with that email address."	Pass
TC_04	(ECP) Verify Email field validation without domain extension.	1. Enter Email: shreyas@gmail2. Password: Pass@1233. Click Login	Error message: "Please enter a valid email address."	Error message: "Please enter a valid email address."	Pass
TC_05	(BVA) Verify Mobile Number field does not accept nonnumeric characters.	1. Enter Mobile: 98765ABCDE 2. Password: Pass@1233. Click Login	Field should not accept characters OR show error: "Please enter a valid number."	Field should not accept characters OR show error: "Please enter a valid number."	Pass

2. Test the system's response for multiple failed login attempts (e.g., lockout feature).

ANS:-

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_06	Verify system response for multiple failed login attempts (lockout feature).	1. Enter Email: shreyas@gmail.com 2. Password: WrongPass13. Click Login4. Repeat steps 1–3 for 5 consecutive times	After 5 failed attempts, system should temporarily lock the account and display error: "Your account has been locked due to multiple failed login attempts. Please try again after 15 minutes."	(To be tested)	(Pass/Fail)



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3. Apply negative testing for empty, whitespace, and invalid credentials.

ANS:-

Test Case ID	Objective	Steps (Input)	Expected Output	Actual Output	Status
TC_01	Verify login with empty Email field.	1. Leave Email empty2. Enter Password: Pass@1233. Click Login	Error message: "Email field cannot be empty."	(To be tested)	(Pass/Fail)
TC_02	Verify login with empty Password field.	1. Enter Email: shreyas@gmail.com 2. Leave Password empty3. Click Login	Error message: "Password field cannot be empty."	(To be tested)	(Pass/Fail)

TC_03	Verify login with whitespace in Email field.	1. Enter Email: " "2. Enter Password: Pass@1233. Click Login	Error message: "Please enter a valid email address."	(To be tested)	(Pass/Fail)
TC_04	Verify login with whitespace in Password field.	1. Enter Email: shreyas@gmail.com 2. Enter Password: " "3. Click Login	Error message: "Password must not be empty or contain only spaces."	(To be tested)	(Pass/Fail)
TC_05	Verify login with invalid Email and Password combination.	1. Enter Email: shreyaswrong@gmail.com 2. Enter Password: WrongPass13. Click Login	Error message: "Invalid email or password."	(To be tested)	(Pass/Fail)

4. Record the expected vs. actual outputs in your test log.

ANS:-

Test Case ID	Test Scenario	Expected Output	Actual Output	Status
TC_01	Mobile Number < 10 digits	Error: "Invalid mobile number"	Error displayed: "Please enter valid number"	Pass
TC_02	Password < 8 characters	Error: "Password too short"	Error displayed correctly	Pass
TC_03	5 Failed	Account Lock or	CAPTCHA	Pass



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Test Case ID	Test Scenario	Expected Output	Actual Output	Status
	Login Attempts	CAPTCHA	appeared after 5th try	

TC_04	Empty Fields Validation	Error: "Enter email/mobile"	Message shown	Pass
TC_05	Whitespace Inputs	Treat as empty / Error	System ignored spaces, showed error	Pass

III. Conclusion:

The login functionality of the e-commerce application was thoroughly tested using **positive and negative scenarios**, including valid credentials, invalid credentials, empty fields, whitespace inputs, password length, and multiple failed attempts. The system correctly handled valid logins by granting access, and provided clear, user-friendly error messages for invalid inputs or failed attempts. Security features like CAPTCHA or account lockout were also triggered appropriately after repeated failed logins, ensuring account safety. Overall, the login process is **robust, secure, and user-friendly**, meeting standard functional and security requirements.

Marks Obtained			Dated Signature of Teacher
Process Related (15)	Product Related (10)	Total (25)	